

MULTI-STAGE AI PROMPT WORKFLOW

Patient Intake Triage Priority Prediction and Escalation

Assessment

Name : Vummadichetty jaya Varshini

Reg No: 99230040812

Course: Gen-AI

College:kalasalingam

1. Scenario Selected

The selected scenario is a Patient Intake Triage Priority Prediction and Escalation workflow. The objective is to transform unstructured patient intake presentations into a structured triage plan, draft an operational response, and then use AI to evaluate and refine the response.

The workflow also considers repeated visits for the same complaint and how long a patient has been waiting. Repeated visits increase the urgency because they can indicate a worsening or undiagnosed condition. If a patient remains unseen beyond the defined time limit, the attending physician should receive an escalation notification.

Note: this workflow is a training exercise in prompt-chain design for an intake queueing system. It is not a clinical decision-support tool and does not replace assessment by a qualified clinician.

2. Objective

- Classify patient intake presentations into suitable triage levels.
- Identify repeated visits and use them as an urgency signal.
- Generate a practical action plan for the intake / triage team.
- Escalate patients who remain unseen beyond the allowed time.
- Use a three-step chained AI workflow with a mandatory self-critique stage.

3. Input Data Used

Patient ID	Chief Complaint / Presentation	Location	Repeat Visit?	Time Waiting
P001	Crushing chest pain radiating to left arm, sweating, short of breath.	Room A	No	10 min
P002	Mild seasonal allergy symptoms, sneezing and itchy eyes.	Room B	No	45 min

Patient ID	Chief Complaint / Presentation	Location	Repeat Visit?	Time Waiting
P003	Child, fever 104°F, lethargic and difficult to rouse.	Room C	Yes -- 2nd visit today	15 min
P004	Sprained ankle after a fall, swelling, able to bear some weight.	Room D	No	40 min
P005	Worsening abdominal pain, reported at three visits this week.	Room E	Yes -- 3	30 min

4. Prompt Chain Transcript and Intermediate AI Responses

Step 1 -- Extraction and Planning

Prompt 1 -- Exact prompt used:

Act as an expert emergency-department triage analyst. I will provide a set of patient intake presentations. Analyze them and extract the key factors that should determine triage priority. Consider safety/life-threat risk, vulnerable patients (e.g. children), repeated visits for the same complaint, symptom severity, and how long the patient has been waiting. Create a structured triage framework and apply it to the patients. Also identify which patients should trigger physician escalation if they remain unseen. Present the result as a table with Patient ID, Triage Level, Reason, Suggested Action, and Escalation Condition. Input patients: P001 chest pain radiating to left arm with sweating; P002 mild seasonal allergy symptoms; P003 child with 104°F fever and lethargy, second visit today; P004 sprained ankle with swelling; P005 worsening abdominal pain reported across three visits this week.

Intermediate AI Response -- Step 1:

ID	Triage Level	Reason	Suggested Action	Escalation Condition
P001	Critical	Cardiac symptom pattern (chest pain, radiation, diaphoresis) suggests possible acute coronary event.	Move to resuscitation bay immediately; obtain ECG and vitals; notify physician now.	Escalate to physician immediately -- no waiting time permitted.
P003	Critical	High fever with lethargy in a child is a red-flag sign, and this is a repeat visit for the same issue.	Priority physician assessment; recheck vitals and hydration status.	Escalate to physician if not seen within 15 minutes.
P005	High	Persistent abdominal pain with repeated visits signals a possible worsening or undiagnosed condition.	Full abdominal assessment; consider imaging or labs; review prior visit notes.	Escalate if unresolved or unseen within 30 minutes.
P004	Medium	Musculoskeletal injury with functional mobility; not an	X-ray if indicated; splint and pain management.	Escalate if unresolved within 90 minutes.

ID	Triage Level	Reason	Suggested Action	Escalation Condition
		immediate threat to life or limb.		
P002	Low	Mild, self-limited allergy symptoms with no safety risk.	Provide symptomatic care or antihistamine guidance.	Escalate only if symptoms worsen or wait exceeds 3 hours.

Step 2 -- Generation / Drafting

Prompt 2 -- Exact prompt used:

Using only the triage framework and patient analysis you produced in Step 1, create an intake response plan for the care team. Do not introduce new triage categories. The plan must explain how patients are handled in priority order, how repeated visits increase urgency, and how patients who remain unseen are escalated. Include a simple notification message that can be sent to the attending physician when an escalation condition is reached. Keep the response practical and suitable for a hospital intake / triage system.

Intermediate AI Response -- Step 2:

The system should process patients in the following order: Critical, High, Medium, and Low. Life-threat presentations receive the highest priority. Patients who have returned repeatedly with the same or worsening complaint should receive increased urgency because repeat visits can indicate a deteriorating condition. Vulnerable patients, such as young children with high fever, are treated with the same urgency as adult life-threat presentations.

For each patient, the system records the intake time, current status, triage level, number of prior related visits, assigned clinician, and time to assessment. The system checks unseen patients against their escalation condition. When the allowed wait time is exceeded, an automatic notification is sent to the attending physician.

Example escalation notification:

ALERT: Patient P003 has exceeded the 15-minute assessment limit. Presentation: high fever with lethargy (pediatric). Repeat visits: 2. Triage Level: CRITICAL. Please assess immediately.

Step 3 -- Evaluation and Refinement (Critique Gate)

Prompt 3 -- Exact prompt used:

Before finalizing the response plan, act as a strict senior reviewer. Evaluate the Step 2 draft against this checklist: (1) Does it clearly prioritize life-threat and vulnerable-patient presentations? (2) Does it explicitly use repeated visits as an urgency factor? (3) Does it explain what happens when a patient is not assessed within the specified time? (4) Does the physician notification contain enough information to act? (5) Is the workflow practical for a hospital intake / triage system? First list any weaknesses you find. Then produce a corrected final version that fixes those weaknesses without changing the core information from Step 1.

Intermediate AI Response -- Step 3 Critique:

- The draft correctly places life-threat and vulnerable-patient presentations ahead of lower-severity complaints.
- Repeated visits are mentioned as an urgency factor, but the final workflow should state this rule more explicitly.
- The escalation process is present, but the final version should clearly state that the system continuously checks unseen patients against their wait-time limits.
- The notification contains the essential presentation information, but it can be made more actionable by including triage level, repeat-visit count, and elapsed wait time.

Refined Final Output:

The proposed patient intake triage workflow first classifies presentations according to life-threat risk, vulnerability of the patient, symptom severity, repeated visits, and time remaining before escalation. A presentation involving an immediate life threat, or a high-risk sign in a vulnerable patient, is treated as Critical. High-severity presentations are treated as High when they affect a patient's safety or are part of a repeated pattern of visits. Medium and Low triage levels are used for presentations with less immediate risk.

Repeated visits are treated as an additional urgency signal. For example, if a patient returns multiple times for the same or a worsening complaint, the system flags the repeated pattern and gives it greater attention than an isolated presentation of the same general type.

After a patient is assigned to a clinician or care area, the system continuously checks whether the patient has been assessed. If the configured wait-time limit is exceeded, the system automatically escalates the case to the attending physician.

Final Physician Notification Example:

ALERT -- Unassessed Patient. Patient ID: P003. Presentation: High fever with lethargy (pediatric). Triage Level: CRITICAL. Prior related visits: 2. Time waiting: 15+ minutes. Status: Unassessed. Action required: Please assess and assign immediate care.

5. Final Workflow Summary

Stage	Input	AI / System Action	Output
1. Extraction / Planning	Raw patient intake presentations	Identify triage factors and classify patients.	Triage framework and patient priority table
2. Generation / Drafting	Step 1 output	Create intake response plan and escalation process.	Operational response plan + physician notification
3. Evaluation / Refinement	Step 2 output	Review against checklist and correct weaknesses.	Final polished workflow

6. Expected System Output

- A ranked list of patients based on triage urgency.
- A triage label such as Critical, High, Medium, or Low.
- Detection of repeated visits for the same or a related complaint.
- Assignment of patients to the responsible clinician or care area.
- Tracking of unassessed patients and elapsed wait time.
- Automatic physician notification when a patient exceeds their escalation limit.
- A dashboard that allows staff to see the most urgent and overdue patients first.

7. Compliance with Assessment Rules

Requirement	Status
At least 3 chained prompts	Satisfied -- three sequential prompts are documented.
Each prompt depends on previous output	Satisfied -- Step 2 uses Step 1; Step 3 reviews and refines Step 2.

Requirement	Status
Critique gate	Satisfied -- Step 3 explicitly evaluates the previous draft against a checklist.
Document exact prompts	Satisfied -- all three prompts are included verbatim in this submission.
Intermediate responses	Satisfied -- outputs from each stage are documented.
Live chat evidence	Screenshot must be inserted in Section 7 from the actual AI interface.

8. Conclusion

This multi-stage AI workflow demonstrates how a patient intake triage problem can be handled through chained prompting rather than a single request. The first stage structures the problem, the second stage converts the analysis into an operational plan, and the third stage critically evaluates and improves the result. The proposed approach also addresses repeated visits and delayed assessment by using them as signals for increased urgency and physician escalation.